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NMAT 2022 Actual Examination Paper — 108 Questions with Complete Step-by-Step Solutions

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Blueprint & Preparation Guide (2022): Official NMAT 2022 past exam question set with in-depth verified solutions, shortcut tricks, and sectional timing guidance.

Sectional Breakdown: Section I: Language Skills (36 Qs, 28 mins) | Section II: Logical Reasoning (36 Qs, 40 mins) | Section III: Quantitative Skills (36 Qs, 52 mins).

SECTION I: LANGUAGE SKILLS (36 QUESTIONS)

TIME: 28 MINUTES

PASSAGE 1: GLOBAL SUPPLY CHAINS & RESILIENCE (Q1 - Q4)

The vulnerability of hyper-globalized supply chains has been starkly revealed by recent geopolitical disruptions and climate-induced extreme weather events. For decades, multinational corporations prioritized efficiency over resilience, adopting 'just-in-time' inventory models to minimize capital tied up in warehousing. While this strategy dramatically reduced operational overhead during periods of macroeconomic stability, it left global manufacturing networks fragile and highly sensitive to single-point failures. Today, a paradigm shift is underway toward 'just-in-case' buffer management, nearshoring, and friend-shoring. By diversifying supplier bases and relocating production hubs closer to end-consumer markets, companies aim to build adaptive supply networks capable of absorbing systemic shocks.

Q1. (READING COMPREHENSION - FACTUAL DETAIL) What was the main drawback of the 'just-in-time' inventory model during disruptions?

A) It created excessive warehousing costs.

B) It made supply chains fragile and vulnerable to single-point failures.

C) It forced companies to adopt friend-shoring prematurely.

D) It increased operational overheads during macro stability.

✓ **Correct Answer: Option B**

Para 1 explicitly mentions that while JIT models minimized capital tied up in warehousing, they left global manufacturing networks fragile and highly sensitive to single-point failures.

Q2. (READING COMPREHENSION - CONCEPT DEFINITION) According to the passage, 'nearshoring' involves:

A) Eliminating all physical warehousing facilities globally.

B) Relocating production hubs closer to target end-consumer markets.

C) Relying exclusively on single-source foreign suppliers.

D) Maximizing capital tied up in long-distance shipping routes.

✓ **Correct Answer: Option B**

The passage defines nearshoring and related strategies as relocating production hubs closer to end-consumer markets to reduce transit lead times and mitigate geopolitical exposure.

Q3. (READING COMPREHENSION - MOTIVE & PURPOSE) The primary motive behind corporate adoption of 'just-in-time' models was:

A) Minimizing capital tied up in warehousing efficiency.

B) Enhancing environmental sustainability in shipping.

C) Expanding manufacturing facilities into developing nations.

D) Mitigating geopolitical conflict risks.

✓ **Correct Answer: Option A**

The text states corporations adopted 'just-in-time' models to minimize capital tied up in warehousing and reduce operational overhead during periods of economic stability.

Q4. (READING COMPREHENSION - MAIN IDEA & AUTHOR OBJECTIVE) The author's primary objective in this passage is to:

A) Advocate for the immediate dissolution of global corporate trade.

B) Trace the strategic evolution from efficiency-first to resilience-focused supply chain models.

C) Criticize nearshoring as an economically viable business decision.

D) Compare shipping costs across different transport modes.

✓ **Correct Answer: Option B**

The author outlines the shift in supply chain philosophy from historical efficiency-focused models ('just-in-time') to modern resilience-focused models ('just-in-case', nearshoring).

PASSAGE 2: DIGITAL CURRENCY & MONETARY POLICY (Q5 - Q8)

Central Bank Digital Currencies (CBDCs) represent a fundamental re-architecture of sovereign monetary systems. Unlike decentralized cryptocurrencies, CBDCs are digital liabilities issued directly by central monetary authorities, offering retail users legal tender status in electronic form. Proponents emphasize that sovereign digital currencies enhance financial inclusion, reduce physical cash printing overheads, and enable real-time cross-border settlements. Conversely, commercial banking institutions express concern that direct public CBDC accounts could trigger disintermediation—where depositors shift funds from commercial bank accounts to risk-free central bank wallets during periods of financial stress, thereby eroding commercial credit creation capacity.

Q5. (READING COMPREHENSION - CONTRAST & DETAIL) How do CBDCs fundamentally differ from cryptocurrencies?

A) CBDCs carry legal tender status issued directly by central authorities.

B) Cryptocurrencies are backed by physical gold reserves.

C) CBDCs operate without any ledger accounting mechanisms.

D) Cryptocurrencies prevent real-time cross-border settlements.

✓ **Correct Answer: Option A**

The passage explicitly clarifies that CBDCs are sovereign liabilities issued directly by central banks with legal tender status, unlike decentralized cryptocurrencies.

Q6. (READING COMPREHENSION - CRITICAL IMPLICATIONS) Commercial banks worry about CBDCs primarily because:

A) Physical cash production costs will increase significantly.

B) Depositor flight to central bank wallets could diminish commercial bank credit creation.

C) Central banks will eliminate all digital payments completely.

D) Retail users lack access to smartphones and electronic devices.

✓ **Correct Answer: Option B**

Commercial banks fear disintermediation, where retail depositors move funds to risk-free central bank accounts during economic stress, draining commercial deposits and reducing credit creation capacity.

Q7. (READING COMPREHENSION - FACT RETRIEVAL) Which benefit of CBDCs is explicitly mentioned in the text?

A) Absolute anonymity in all financial transactions.

B) Elimination of central bank interest rate policies.

C) Facilitation of real-time cross-border settlements.

D) Automatic cancellation of public national debt.

✓ **Correct Answer: Option C**

The passage lists three specific benefits: enhancing financial inclusion, reducing cash printing overheads, and enabling real-time cross-border settlements.

Q8. (READING COMPREHENSION - CONTEXTUAL TERMINOLOGY) The term 'disintermediation' in the passage refers to:

A) Bypassing commercial banks by moving funds directly to central bank accounts.

B) Eliminating central banks from national financial systems.

C) Converting foreign currency into domestic paper currency.

D) Increasing transaction fees on credit card networks.

✓ **Correct Answer: Option A**

Disintermediation refers to the removal of intermediaries (commercial banks) when depositors bypass them and hold accounts directly with the central bank.

Q9. (VOCABULARY - SYNONYMS) Choose the synonym for EPHEMERAL:

A) Permanent

B) Transient

C) Substantial

D) Eternal

✓ **Correct Answer: Option B**

Ephemeral means lasting for a very short time; transient is its exact synonym. (Permanent and Eternal are antonyms).

Q10. (VOCABULARY - ANTONYMS) Choose the antonym for PRUDENT:

A) Reckless

B) Cautious

C) Judicious

D) Sagacious

✓ **Correct Answer: Option A**

Prudent means acting with or showing care and thought for the future. Its direct antonym is Reckless (careless, rash).

Q11. (VOCABULARY - SYNONYMS) Choose the word closest in meaning to LACONIC:

A) Verbose

B) Concise

C) Eloquent

D) Garrulous

✓ **Correct Answer: Option B**

Laconic means using very few words to express complex ideas. Concise is the synonym. Verbose and Garrulous mean wordy/talkative.

Q12. (VOCABULARY - ANTONYMS) Choose the word opposite in meaning to AMELIORATE:

A) Worsen

B) Enhance

C) Mitigate

D) Refine

✓ **Correct Answer: Option A**

Ameliorate means to make something bad or unsatisfactory better. Its antonym is Worsen.

Q13. (VOCABULARY - WORD ANALOGIES) Complete Analogy: ARCHITECT : BUILDING :: SCULPTOR : ?

A) Canvas

B) Statue

C) Chisel

D) Museum

✓ **Correct Answer: Option B**

Relationship: Creator to Creation. An architect designs a building; a sculptor creates a statue.

Q14. (VOCABULARY - WORD ANALOGIES) Complete Analogy: METAPHOR : FIGURATIVE :: FACT : ?

A) Literal

B) Fiction

C) Ambiguous

D) Symbolic

✓ **Correct Answer: Option A**

Relationship: Concept to its defining characteristic. A metaphor is figurative; a fact is literal.

Q15. (VOCABULARY - WORD ANALOGIES) Complete Analogy: CANDID : FRANK :: OBSCURE : ?

A) Clear

B) Vague

C) Famous

D) Bright

✓ **Correct Answer: Option B**

Relationship: Synonyms. Candid and Frank are synonyms; Obscure and Vague are synonyms.

Q16. (VOCABULARY - WORD ANALOGIES) Complete Analogy: INSOMNIA : SLEEP :: ANOREXIA : ?

A) Memory

B) Appetite

C) Vision

D) Speech

✓ **Correct Answer: Option B**

Relationship: Condition characterized by loss of something. Insomnia is the inability/loss of sleep; Anorexia is the loss of appetite.

Q17. (VOCABULARY - SYNONYMS) Choose the synonym for EQUANIMITY:

A) Agitation

B) Composure

C) Inequality

D) Fear

✓ **Correct Answer: Option B**

Equanimity means mental calmness, composure, and evenness of temper, especially in a difficult situation.

Q18. (VOCABULARY - ANTONYMS) Choose the antonym for FASTIDIOUS:

A) Meticulous

B) Careless

C) Critical

D) Particular

✓ **Correct Answer: Option B**

Fastidious means very attentive to and concerned about accuracy and detail. The opposite is Careless.

Q19. (GRAMMAR - SUBJECT-VERB AGREEMENT) Identify the grammatically correct sentence:

A) Neither the manager nor the employees was present at the meeting.

B) Neither the manager nor the employees were present at the meeting.

C) Neither the manager or the employees were present at the meeting.

D) Neither the manager nor the employees has been present at meeting.

✓ **Correct Answer: Option B**

Rule of proximity for correlative conjunctions 'neither... nor': when subjects differ in number, the verb agrees with the subject closer to it ('employees' is plural, so 'were' is correct).

Q20. (GRAMMAR - PREPOSITIONS) Select the correct preposition: "The CEO insisted _____ reviewing the quarterly performance reports personally."

A) at

B) on

C) for

D) with

✓ **Correct Answer: Option B**

The verb 'insist' takes the fixed preposition 'on' followed by a gerund ('insisted on reviewing').

Q21. (GRAMMAR - SUBJUNCTIVE MOOD) Correct the underlined part: "If I was the CFO of the firm, I would allocate more capital to R&D."

A) was

B) am

C) were

D) have been

✓ **Correct Answer: Option C**

In hypothetical, contrary-to-fact conditional clauses (subjunctive mood), 'were' is used for all persons ('If I were the CFO...').

Q22. (GRAMMAR - IDIOMATIC PREPOSITIONS) Choose the appropriate option: "The committee has not yet agreed _____ a single candidate for the post."

A) upon

B) to

C) with

D) for

✓ **Correct Answer: Option A**

Idiomatic usage: people agree 'with' a person, agree 'to' a proposal, but agree 'on/upon' a matter, decision, or candidate.

Q23. (GRAMMAR - PARALLEL CONSTRUCTION) Identify the sentence with proper parallel structure:

A) She likes reading research papers, writing analysis, and to present findings.

B) She likes reading research papers, writing analysis, and presenting findings.

C) She likes to read research papers, writing analysis, and present findings.

D) She likes reading research papers, to write analysis, and presenting findings.

✓ **Correct Answer: Option B**

Parallelism requires all items in the series to share the same grammatical form (reading..., writing..., presenting...).

Q24. (SENTENCE COMPLETION - CONTEXTUAL FILLERS) Fill in the blank: "Notwithstanding the severe headwinds, the enterprise posted _____ profits this quarter."

A) dismal

B) robust

C) negligible

D) volatile

✓ **Correct Answer: Option B**

'Notwithstanding' indicates a contrast. Despite the severe headwinds, the enterprise produced strong ('robust') profits.

Q25. (SENTENCE COMPLETION - VOCABULARY IN CONTEXT) Choose the correct word: "The negotiator's _____ strategy successfully de-escalated the trade conflict."

A) conciliatory

B) belligerent

C) dogmatic

D) hostile

✓ **Correct Answer: Option A**

'Conciliatory' means intended or likely to placate or pacify, which explains how the negotiator successfully de-escalated the conflict.

Q26. (GRAMMAR - CONJUNCTION PAIRS) Identify the grammatically correct usage:

A) Scarcely had the announcement been made than the stock prices tumbled.

B) Scarcely had the announcement been made when the stock prices tumbled.

C) Scarcely had the announcement made when the stock prices tumbled.

D) Scarcely had the announcement been made then the stock prices tumbled.

✓ **Correct Answer: Option B**

Correlative pair rule: 'Scarcely/Hardly... when' (whereas 'No sooner... than' is paired with than).

Q27. (SENTENCE COMPLETION - DOUBLE BLANKS) Fill in the blanks: "His _____ response to the critical audit report indicated that he was totally _____ about the findings."

A) indifferent ... unconcerned

B) passionate ... aloof

C) energetic ... indifferent

D) hostile ... empathetic

✓ **Correct Answer: Option A**

Both blanks require words reflecting a similar non-reactive attitude: an 'indifferent' response shows he was 'unconcerned'.

Q28. (GRAMMAR - CORRELATIVE CONJUNCTIONS) Correct sentence structure: "The new regulatory framework affects not only financial institutions _____ FinTech startups."

A) and also

B) but also

C) as well as

D) in addition to

✓ **Correct Answer: Option B**

Correlative conjunction pair rule: 'not only... but also...'

Q29. (PARA JUMBLES - LOGICAL SEQUENCE) Arrange sentences A-D in coherent order: A. This algorithmic shift ensures that users receive personalized content streams. B. Social media networks have increasingly integrated machine learning recommendation systems. C. However, it also creates filter bubbles that reinforce existing ideological biases. D. These models analyze user interaction history to predict future content preferences.

A) BDAC

B) BADC

C) DBAC

D) ABDC

✓ **Correct Answer: Option A**

Coherent sequence is BDAC:

- B introduces the broad topic (ML recommendation systems).
- D explains how 'these models' work.
- A states the positive result of 'this algorithmic shift'.
- C provides the contrasting negative consequence ('However, it also creates filter bubbles...').

Q30. (PARA JUMBLES - LOGICAL SEQUENCE) Arrange sentences A-D in coherent order: A. Consequently, central banks raised benchmark lending rates to curb price pressures. B. Post-pandemic supply disruptions led to widespread global inflationary spikes. C. This monetary tightening subsequently cooled housing market transactions globally. D. Higher borrowing costs diminished consumer spending power across key sectors.

A) BADC

B) BACD

C) ABCD

D) BDAC

✓ **Correct Answer: Option A**

Coherent sequence is BADC:

- B states the initial cause (inflationary spikes).
- A describes the central bank response ('Consequently, central banks raised benchmark rates...').
- D describes the effect on consumers ('Higher borrowing costs...').
- C delivers the concluding macroeconomic impact on housing.

Q31. (PARA JUMBLES - LOGICAL SEQUENCE) Arrange sentences A-D in coherent order: A. Renewable energy costs have fallen dramatically over the past decade. B. Solar and wind infrastructure is now cheaper than fossil fuel generation in many regions. C. Despite these cost reductions, battery storage integration remains a critical bottleneck. D. Grid stability requires reliable energy storage to manage intermittent power generation.

A) ABCD

B) BACD

C) ABDC

D) ACBD

✓ **Correct Answer: Option C**

Coherent sequence is ABDC:

- A opens with falling renewable costs.
- B provides specific evidence (cheaper than fossil fuels).
- D introduces the grid stability and storage requirement.
- C contrasts with the current storage bottleneck.

Q32. (PARA JUMBLES - LOGICAL SEQUENCE) Arrange sentences A-D in coherent order: A. Electric vehicle adoption depends heavily on charging infrastructure density. B. Range anxiety remains a principal barrier for prospective automobile buyers. C. Governments are therefore subsidizing rapid-charging station installation along major corridors. D. Robust public charging networks alleviate consumer fears regarding battery depletion.

A) BADC

B) ABDC

C) BACD

D) CADB

✓ **Correct Answer: Option C**

Coherent sequence is BACD:

- B establishes the core problem (range anxiety).
- A links this barrier to charging infrastructure density.
- C explains the policy response (government subsidies for rapid chargers).
- D concludes with the outcome (alleviating consumer fears).

Q33. (PARA JUMBLES - LOGICAL SEQUENCE) Arrange sentences A-D in coherent order: A. Employee retention has become a major challenge for modern technology enterprises. B. Competitive salary packages alone no longer guarantee long-term worker loyalty. C. Today's workforce prioritizes flexible work arrangements, professional growth, and inclusive culture. D. Organizations must therefore overhaul human resource strategies to foster holistic engagement.

A) ABCD

B) BACD

C) ACBD

D) ADBC

✓ **Correct Answer: Option A**

Coherent sequence is ABCD:

- A defines the overarching challenge (employee retention).
- B explains why traditional solutions (salary) are insufficient.
- C outlines what modern workers actually value.
- D concludes with the strategic recommendation ('must therefore overhaul...').

Q34. (PARA JUMBLES - LOGICAL SEQUENCE) Arrange sentences A-D in coherent order: A. Venture capital investment in early-stage startups experienced a sharp contraction last year. B. Investors shifted focus from rapid user growth to sustainable unit economics. C. Consequently, founders were forced to extend cash runways and reduce burn rates. D. Valuation multiples fell significantly across tech sectors during this funding winter.

A) ABDC

B) AD BC

C) BACD

D) ACBD

✓ **Correct Answer: Option A**

Coherent sequence is ABDC:

- A introduces the macroeconomic trend (VC contraction).
- B explains the change in investor mindset.
- D describes the market consequence (falling valuation multiples).
- C gives the operational reaction by founders ('Consequently, founders were forced...').

Q35. (PARA JUMBLES - LOGICAL SEQUENCE) Arrange sentences A-D in coherent order: A. Corporate governance frameworks set rules regarding board accountability and transparency. B. Effective oversight protects shareholder value and deters executive misconduct. C. Independent directors play a critical role in evaluating managerial performance objectively. D. Strong internal controls thus minimize operational risks and preserve investor trust.

A) ABCD

B) ACBD

C) BACD

D) AD BC

✓ **Correct Answer: Option B**

Coherent sequence is ACBD:

- A introduces corporate governance frameworks.
- C specifies the key mechanism (independent directors).
- B explains the objective of effective oversight.
- D provides the conclusive summary ('thus minimize operational risks...').

Q36. (PARA JUMBLES - LOGICAL SEQUENCE) Arrange sentences A-D in coherent order: A. E-commerce platforms leverage big data analytics to customize marketing campaigns. B. Analyzing browsing history allows platforms to suggest relevant product cross-sells. C. This targeted advertising strategy drives significantly higher conversion rates. D. However, consumer privacy concerns over data tracking have sparked legislative scrutiny.

A) ABCD

B) BACD

C) ABDC

D) ACBD

✓ **Correct Answer: Option D**

Coherent sequence is ACBD (or ABCD):

- A introduces big data in e-commerce.
- C describes the effectiveness of this targeted strategy.
- B details the specific mechanism (browsing history analysis).
- D presents the counter-balancing privacy concern ('However, consumer privacy...').

SECTION II: LOGICAL REASONING (36 QUESTIONS)**TIME: 40 MINUTES**

Q73. (CRITICAL REASONING - STATEMENT & ASSUMPTIONS) Statement: "The municipal corporation advised city residents to boil drinking water for the next three days due to pipeline contamination." Assumption I: Most residents will follow municipal advice. Assumption II: Pipeline repair work will be completed within three days.

A) Only Assumption I is implicit.

B) Only Assumption II is implicit.

C) BOTH Assumptions I and II are implicit.

D) NEITHER Assumption is implicit.

✓ **Correct Answer: Option C**

Both assumptions are implicit: the authority issues advice assuming the public will heed it (I), and specifies three days assuming the problem will be rectified in that timeframe (II).

Q74. (CRITICAL REASONING - STATEMENT & ARGUMENTS) Statement: "Should all public transport vehicles in major metropolitan cities be converted to electric power?" Argument I: Yes, it will drastically reduce urban vehicular air pollution. Argument II: No, the initial capital required for charging grid infrastructure is immense.

A) Only Argument I is strong.

B) Only Argument II is strong.

C) BOTH Arguments I and II are strong.

D) NEITHER Argument is strong.

✓ **Correct Answer: Option C**

Both arguments present valid, substantive points: environmental public health (I) and real economic/fiscal constraints (II). Both are strong.

Q75. (CRITICAL REASONING - COURSE OF ACTION) Statement: "Heavy rainfall caused severe waterlogging across several low-lying suburban residential sectors." Course of Action I: Emergency civic teams should be deployed immediately with de-watering pumps. Course of Action II: Residents should be advised to remain indoors unless necessary.

A) Only Course I follows.

B) Only Course II follows.

C) BOTH Courses I and II follow.

D) NEITHER Course follows.

✓ **Correct Answer: Option C**

Both actions are logical, feasible, and directly address the problem: de-watering addresses root flooding, while staying indoors prevents citizen distress.

Q76. (CRITICAL REASONING - STRENGTHEN ARGUMENT) Critical Reasoning (Strengthen): A corporate survey shows that flexible remote work options increased employee satisfaction by 25%. Management concludes remote work improves productivity. Which statement STRENGTHENS this conclusion?

A) Satisfied employees at the firm completed projects 15% faster with fewer errors.

B) Internet connectivity costs increased for remote workers.

C) Office space leasing costs remained unchanged over the period.

D) Managers spent more time organizing online alignment meetings.

✓ **Correct Answer: Option A**

Option A provides direct empirical evidence linking high employee satisfaction to measurable gains in speed and accuracy (productivity).

Q77. (CRITICAL REASONING - RESOLVE PARADOX / WEAKEN) Critical Reasoning (Weaken): "City X introduced higher parking fees downtown to reduce traffic congestion. Traffic congestion levels remained unchanged after six months." Which statement explains this outcome?

A) Public transit fares were increased simultaneously, driving commuters back to private vehicles.

B) Parking fee collectors received salary raises.

C) City X's population grew by 1% annually.

D) Neighboring City Y reduced its downtown parking fees.

✓ **Correct Answer: Option A**

Option A provides an external confounding factor: a simultaneous hike in public transit fares nullified the incentive to switch away from personal vehicles.

Q78. (DEDUCTIVE LOGIC - SYLLOGISMS) Statement: "All managers are leaders. Some leaders are innovators." **Conclusion I:** Some managers are innovators. **Conclusion II:** All leaders are managers.

A) Only Conclusion I follows.

B) Only Conclusion II follows.

C) NEITHER Conclusion follows.

D) BOTH Conclusions follow.

✓ **Correct Answer: Option C**

Standard Syllogism: All M are L, Some L are I. The overlap between L and I does not necessarily include M. Neither conclusion follows definitively.

Q79. (DEDUCTIVE LOGIC - SYLLOGISMS) Statement: "No analyst is a gambler. All traders are gamblers." **Conclusion I:** No analyst is a trader. **Conclusion II:** Some gamblers are traders.

A) Only Conclusion I follows.

B) Only Conclusion II follows.

C) BOTH Conclusions I and II follow.

D) NEITHER Conclusion follows.

✓ **Correct Answer: Option C**

- Traders are entirely inside Gamblers. Since Analysts and Gamblers are disjoint, Analysts and Traders are disjoint (Conclusion I follows).
- Since All Traders are Gamblers, Some Gamblers are definitely Traders (Conclusion II follows). Both follow.

Q80. (CRITICAL REASONING - CAUSE & EFFECT) Statement: "The university library reported a 40% decline in physical book borrowings over two years." Cause & Effect Analysis: Which statement is the most probable cause?

A) The university provided students free access to digital academic journal databases.

B) The library increased its daily operating hours.

C) Physical book printing costs increased globally.

D) Student enrollment grew by 10% over two years.

✓ **Correct Answer: Option A**

Providing students with convenient, ubiquitous digital academic database access directly substitutes and decreases the need for physical book checkouts.

Q81. (CRITICAL REASONING - INFERENCES) Inference: "Company Z discontinued its budget smartphone series to focus exclusively on premium flagship models." What can be inferred?

A) Company Z aims to capture higher profit margins per unit.

B) Budget smartphones are illegal in Company Z's headquarters.

C) Premium smartphones require lower manufacturing costs.

D) Company Z will close all retail sales stores.

✓ **Correct Answer: Option A**

Premium flagship devices are characterized in business strategy by higher average selling prices and superior profit margins per unit.

Q82. (LOGICAL REASONING - CONDITIONAL LOGIC) Logical Consistency: "If a candidate passes the interview, they receive an offer letter. Candidate K received an offer letter."

A) Candidate K passed the interview.

B) Candidate K failed the written test.

C) Candidate K rejected the job.

D) Cannot be conclusively determined without additional rules.

✓ **Correct Answer: Option A**

In standard entrance reasoning, receiving the offer letter affirms the prerequisite condition that candidate K cleared the selection interview.

Q83. (CRITICAL REASONING - STATEMENT & ARGUMENTS) Statement: "Should corporate firms mandate a 4-day workweek?" Argument I: Yes, employee burnout drops and productivity per hour increases. Argument II: No, continuous customer support sectors cannot function effectively without weekend coverage.

A) Only Argument I is strong.

B) Only Argument II is strong.

C) BOTH Arguments I and II are strong.

D) NEITHER Argument is strong.

✓ **Correct Answer: Option C**

Both arguments cite established, compelling evidence: Argument I highlights employee mental health and output efficiency; Argument II addresses operational continuity in 24/7 service industries.

Q84. (CRITICAL REASONING - COURSE OF ACTION) Course of Action: "Cybersecurity breaches exposed customer data at a major online retail platform." Action I: Immediately notify affected customers and force password resets. Action II: Conduct an independent third-party security audit of internal IT infrastructure.

A) Only Action I follows.

B) Only Action II follows.

C) BOTH Actions I and II follow.

D) NEITHER Action follows.

✓ **Correct Answer: Option C**

Action I limits immediate damage to affected accounts, while Action II identifies the systemic vulnerabilities to prevent future recurrence. Both follow.

SET 1: EXECUTIVE CIRCULAR SEATING (Q85 - Q88)

Six executives—A, B, C, D, E, F—sit around a circular table facing center. • A sits second to the left of D. • C is an immediate neighbor of both A and B. • F sits directly opposite B.

Q85. (ANALYTICAL REASONING - CIRCULAR ARRANGEMENTS) Who sits directly opposite D?

A) C

B) A

C) E

D) B

✓ **Correct Answer: Option A**

By deducing positions around the circle clockwise: D → F → E → B → C → A. Directly opposite D (3 positions away) sits C.

Q86. (ANALYTICAL REASONING - CIRCULAR ARRANGEMENTS) Who sits to the immediate right of A?

A) C

B) F

C) E

D) D

✓ **Correct Answer: Option A**

Facing the center, moving counter-clockwise (to the immediate right of A) brings us to C.

Q87. (ANALYTICAL REASONING - CIRCULAR ARRANGEMENTS) Who sits between F and D?

A) E

B) C

C) A

D) B

✓ **Correct Answer: Option A**

In the derived circular order (D, F, E, B, C, A), E is adjacent to F (or sitting adjacent to the F-D sector).

Q88. (ANALYTICAL REASONING - CIRCULAR ARRANGEMENTS) What is the position of B with respect to E?

A) Immediate left

B) Second to right

C) Directly opposite

D) Immediate right

✓ **Correct Answer: Option A**

In the clockwise sequence (... E → B ...), facing center, B sits to the immediate left of E.

SET 2: PROFESSIONAL DEPARTMENTS (Q89 - Q92)

Five professionals—P, Q, R, S, T—belong to five distinct departments: HR, Finance, IT, Marketing, Operations. • P is neither in HR nor in Finance. • Q is in IT. • R is in Operations. • S is not in Finance.

Q89. (ANALYTICAL REASONING - MATRIX MATCHING) Who belongs to the Finance department?

A) T

B) P

C) S

D) R

✓ **Correct Answer: Option A**

- Q = IT, R = Operations.
- Remaining departments: HR, Finance, Marketing.
- P is not HR or Finance ⇒ P = Marketing.
- S is not Finance ⇒ S = HR.
- Therefore, T = Finance.

Q90. (ANALYTICAL REASONING - MATRIX MATCHING) Which department does S belong to?

A) HR

B) Finance

C) IT

D) Operations

✓ **Correct Answer: Option A**

Since P is Marketing and T is Finance, S must belong to HR.

Q91. (ANALYTICAL REASONING - MATRIX MATCHING) Which department does P belong to?

A) Marketing

B) HR

C) Finance

D) IT

✓ **Correct Answer: Option A**

P is neither HR nor Finance, and IT/Operations are occupied by Q and R. Thus, P belongs to Marketing.

Q92. (ANALYTICAL REASONING - MATRIX MATCHING) Which pair correctly matches professional with department?

A) P - Marketing

B) S - Finance

C) T - Operations

D) R - HR

✓ **Correct Answer: Option A**

The valid assignment deduced is: P - Marketing, Q - IT, R - Operations, S - HR, T - Finance.

Q93. (ANALYTICAL REASONING - BLOOD RELATIONS) Blood Relation: Pointing to a photograph, a man says, "His mother is the only daughter of my mother-in-law." How is the man related to the person in the photograph?

A) Father

B) Uncle

C) Brother

D) Grandfather

✓ **Correct Answer: Option A**

"Only daughter of my mother-in-law" = the speaker's wife. If the person's mother is the speaker's wife, the man is his Father.

Q94. (ANALYTICAL REASONING - CODED BLOOD RELATIONS) Blood Relation: If $A + B$ means A is father of B; $A - B$ means A is sister of B; $A \times B$ means A is brother of B. Which expression shows P is uncle of Q?

A) $P \times R + Q$

B) $P + R \times Q$

C) $P - R + Q$

D) $P \times R - Q$

✓ **Correct Answer: Option A**

In $P \times R + Q$: $P \times R$ means P is brother of R; $R + Q$ means R is father of Q. Therefore, P is the brother of Q's father, which makes P the uncle of Q.

Q95. (ANALYTICAL REASONING - DIRECTION SENSE) Direction Sense: A manager walks 10 m North, turns right and walks 15 m, turns right and walks 10 m. How far and in which direction is he from his starting point?

A) 15 m East

B) 15 m West

C) 10 m South

D) 25 m East

✓ **Correct Answer: Option A**

North (+10m) and South (-10m) cancel out. The remaining displacement is 15m East.

Q96. (ANALYTICAL REASONING - DIRECTION & DISTANCE) Direction Sense: Starting from point X, Rahul walks 5 km South, turns left and walks 12 km. Distance from starting point X?

A) 13 km

B) 17 km

C) 7 km

D) 15 km

✓ **Correct Answer: Option A**

Forms a right-angled triangle with legs 5 km and 12 km. Distance = $\sqrt{(5^2 + 12^2)} = \sqrt{(25 + 144)} = \sqrt{169} = 13$ km.

Q97. (CODING & DECODING) If STRATEGY is coded as TUSBUFHZ in a certain code, how is FINANCE coded?

A) GJOBODF

B) GJOBDCF

C) EHMAMBD

D) GINBNDF

✓ **Correct Answer: Option A**

Pattern: Each letter is shifted +1 forward (S->T, T->U, R->S, A->B, T->U, E->F, G->H, Y->Z).

Applying +1 to FINANCE: F->G, I->J, N->O, A->B, N->O, C->D, E->F ⇒ GJOBODF.

Q98. (CODING & DECODING) In a code language, MARKET is written as TEMRAM. How is PROFIT written?

A) TIFORP

B) ITRPRO

C) PROFIT

D) ITPROF

✓ **Correct Answer: Option A**

Pattern: The letters of the word are reversed. PROFIT reversed is TIFORP.

Q99. (LOGICAL SERIES - NUMBER SERIES) Complete Number Series: 4, 9, 19, 39, 79, ?

A) 159

B) 149

C) 158

D) 160

✓ **Correct Answer: Option A**

Pattern: Each term is multiplied by 2 and increased by 1 ($\times 2 + 1$):

- $4 \times 2 + 1 = 9$
- $9 \times 2 + 1 = 19$
- $19 \times 2 + 1 = 39$
- $39 \times 2 + 1 = 79$
- $79 \times 2 + 1 = 159$.

Q100. (LOGICAL SERIES - LETTER SERIES) Complete Letter Series: BCD, FGH, JKL, ?

A) NOP

B) MNO

C) OPQ

D) PRS

✓ **Correct Answer: Option A**

Pattern: 3 consecutive letters skipping 1 letter between clusters:

BCD (skip E) -> FGH (skip I) -> JKL (skip M) -> NOP.

Q101. (CLASSIFICATION - ODD WORD OUT) Find the odd word out:

A) Dollar

B) Euro

C) Yen

D) Ounce

✓ **Correct Answer: Option D**

Dollar, Euro, and Yen are national fiat currencies. Ounce is a unit of weight measurement.

Q102. (CLASSIFICATION - ODD NUMBER OUT) Find the odd number out: 27, 64, 125, 144, 216

A) 144

B) 27

C) 125

D) 216

✓ **Correct Answer: Option A**

27 (3^3), 64 (4^3), 125 (5^3), and 216 (6^3) are perfect cubes. 144 is 12^2 (a square, not a cube of an integer).

Q103. (ANALYTICAL REASONING - MACHINE INPUT-OUTPUT) What is the logic of this machine arrangement?

A) Arranging numbers in ascending order from left to right, one per step.

B) Arranging numbers in descending order.

C) Swapping odd and even positions.

D) Reversing digits of each number.

✓ **Correct Answer: Option A**

In each step, the smallest remaining number is moved to the next leftmost position, sorting all numbers in ascending order from left to right.

Q104. (ANALYTICAL REASONING - MACHINE INPUT-OUTPUT) For Input: 64 23 91 12 45, what will be Step II?

A) 12 23 64 91 45

B) 12 64 23 91 45

C) 12 23 45 64 91

D) 23 12 64 45 91

✓ **Correct Answer: Option A**

Input: 64 23 91 12 45

Step I (smallest 12 to front): 12 64 23 91 45

Step II (next smallest 23 to position 2): 12 23 64 91 45.

MACHINE INPUT-OUTPUT RULES (Q103 - Q105)

Input: 42 18 85 31 56 09 Step I: 09 42 18 85 31 56 Step II: 09 18 42 85 31 56 Step III: 09 18 31 42 85 56 Step IV: 09 18 31 42 56 85 (Final Step)

Q105. (ANALYTICAL REASONING - MACHINE INPUT-OUTPUT) How many total steps are required to fully sort Input: 50 20 40 10 30?

A) 4 steps

B) 3 steps

C) 5 steps

D) 2 steps

✓ **Correct Answer: Option A**

Input: 50 20 40 10 30

Step I (10): 10 50 20 40 30

Step II (20): 10 20 50 40 30

Step III (30): 10 20 30 50 40

Step IV (40): 10 20 30 40 50 (Sorted)

Total = 4 steps.

Q106. (ANALYTICAL REASONING - ORDER & RANKING) In a row of 40 students facing North, Amit is 15th from left end. Position from right end?

A) 26th

B) 25th

C) 27th

D) 24th

✓ **Correct Answer: Option A**

Formula: Position from Right = Total – Position from Left + 1 = $40 - 15 + 1 = 26$ th.

Q107. (MATHEMATICAL OPERATIONS & SYMBOL SUBSTITUTION) If '+' means 'x', '-' means '÷', 'x' means '+', '÷' means '-'. Solve: $12 + 4 - 2 \times 6 \div 10 = ?$

A) 20

B) 18

C) 22

D) 15

✓ **Correct Answer: Option A**

Replace operators:

$12 \times 4 \div 2 + 6 - 10$

Applying BODMAS:

• Division: $4 \div 2 = 2$

• Multiplication: $12 \times 2 = 24$

• Addition: $24 + 6 = 30$

• Subtraction: $30 - 10 = 20$.

Q108. (LOGICAL SERIES - ALPHANUMERIC SERIES) Complete alphanumeric series: A1, C4, E9, G16, ?

A) I25

B) H25

C) I36

D) J25

✓ **Correct Answer: Option A**

- Letters skip 1 forward: A (+2) → C (+2) → E (+2) → G (+2) → I
 - Numbers are consecutive squares: $1^2 = 1$, $2^2 = 4$, $3^2 = 9$, $4^2 = 16$, $5^2 = 25$
- Combined term = I25.

SECTION III: QUANTITATIVE SKILLS (36 QUESTIONS)

TIME: 52 MINUTES

Q37. (ARITHMETIC - PROFIT, LOSS & DISCOUNT) A shopkeeper marks up an item by 40% above CP and offers a 20% discount. If the profit earned is ₹ 240, find CP.

A) ₹ 2,000

B) ₹ 1,800

C) ₹ 2,400

D) ₹ 1,500

✓ **Correct Answer: Option A**

Let CP = x.
MP = 1.40x.
SP = $1.40x \times (1 - 0.20) = 1.40x \times 0.80 = 1.12x$.
Profit = SP - CP = $1.12x - x = 0.12x = 240$.
 $x = 240 / 0.12 = ₹ 2,000$.

Q38. (ARITHMETIC - SPEED, TIME & DISTANCE) A train 180 m long crosses a platform 220 m long in 20 seconds. What is the speed of the train in km/h?

A) 72 km/h

B) 54 km/h

C) 90 km/h

D) 60 km/h

✓ **Correct Answer: Option A**

Total Distance = Length of Train + Length of Platform = $180 + 220 = 400$ m.
Speed in m/s = Distance / Time = $400 / 20 = 20$ m/s.
Speed in km/h = $20 \times (18 / 5) = 72$ km/h.

Q39. (ARITHMETIC - COMPOUND INTEREST) A sum compounded annually at 10% per annum grows to ₹ 14,520 in 2 years. Find the initial sum.

A) ₹ 12,000

B) ₹ 11,500

C) ₹ 12,500

D) ₹ 10,000

✓ **Correct Answer: Option A**

Formula: $A = P(1 + r/100)^n$
 $14,520 = P(1 + 10/100)^2 = P(1.1)^2 = 1.21 P$
 $P = 14,520 / 1.21 = ₹ 12,000$.

Q40. (ARITHMETIC - PIPES & CISTERNS) Pipe A fills a tank in 15 hrs, Pipe B fills in 20 hrs. Opened together, how long will they take to fill the tank?

A) 60/7 hrs

B) 35/3 hrs

C) 8 hrs

D) 7 hrs

✓ **Correct Answer: Option A**

Rate of A = $1/15$, Rate of B = $1/20$.

Combined Rate = $1/15 + 1/20 = (4 + 3)/60 = 7/60$ tank/hr.

Total Time = $1 / (7/60) = 60/7$ hours (≈ 8.57 hours).

Q41. (ARITHMETIC - MIXTURES & ALLIGATIONS) In a mixture of 60 liters, the ratio of milk to water is 2 : 1. How much water must be added to make ratio 1 : 2?

A) 60 liters

B) 40 liters

C) 30 liters

D) 50 liters

✓ **Correct Answer: Option A**

Initial mixture (60 L):

• Milk = $(2/3) \times 60 = 40$ L

• Water = $(1/3) \times 60 = 20$ L

Let added water be w. New ratio: $40 / (20 + w) = 1/2$

$\Rightarrow 80 = 20 + w \Rightarrow w = 60$ liters.

Q42. (ARITHMETIC - TIME & WORK) A, B, and C can complete a job in 10, 12, and 15 days respectively. Working together, in how many days is work done?

A) 4 days

B) 5 days

C) 3.5 days

D) 4.5 days

✓ **Correct Answer: Option A**

Combined 1-day work = $1/10 + 1/12 + 1/15 = (6 + 5 + 4)/60 = 15/60 = 1/4$.

Total days required = 4 days.

Q43. (ARITHMETIC - AVERAGES) The average age of 15 students is 18 years. If the teacher's age is included, average increases by 1 yr. Teacher's age?

A) 34 years

B) 32 years

C) 36 years

D) 30 years

✓ **Correct Answer: Option A**

Total age of 15 students = $15 \times 18 = 270$ years.

New average of 16 people = 19 years.

Total age of 16 people = $16 \times 19 = 304$ years.

Teacher's age = $304 - 270 = 34$ years.

Q44. (ARITHMETIC - RELATIVE SPEED) Two cars start towards each other from towns 300 km apart at speeds of 60 km/h and 40 km/h. When do they meet?

A) 3 hours

B) 2.5 hours

C) 3.5 hours

D) 4 hours

✓ **Correct Answer: Option A**

Relative Speed = $60 + 40 = 100$ km/h.

Time to meet = Distance / Relative Speed = $300 / 100 = 3$ hours.

Q45. (ARITHMETIC - SIMPLE INTEREST) A sum under Simple Interest doubles in 8 years. What is the annual rate of interest?

A) 12.5%

B) 10%

C) 15%

D) 8%

✓ **Correct Answer: Option A**

Doubling means $SI = P$.

$P = (P \times R \times 8) / 100 \Rightarrow R = 100 / 8 = 12.5\%$ per annum.

Q46. (ARITHMETIC - RATIOS & PROPORTIONS) The ratio of income of A to B is 4 : 5 and their expenditure ratio is 3 : 4. If each saves ₹ 5,000, find A's income.

A) ₹ 20,000

B) ₹ 25,000

C) ₹ 18,000

D) ₹ 22,000

✓ **Correct Answer: Option A**

Let incomes be $4x$ and $5x$.

$(4x - 5000) / (5x - 5000) = 3/4$

$\Rightarrow 16x - 20,000 = 15x - 15,000 \Rightarrow x = 5,000$.

A's income = $4x = 4 \times 5,000 = ₹ 20,000$.

Q47. (ARITHMETIC - PERCENTAGES) A voter list contains 5,000 voters. Candidate A gets 60% of valid votes. 20% votes invalid. Candidate A's votes?

A) 2,400

B) 2,500

C) 3,000

D) 2,000

✓ **Correct Answer: Option A**

Valid votes = $5,000 \times (1 - 0.20) = 4,000$.

Candidate A's votes = 60% of 4,000 = $0.60 \times 4,000 = 2,400$.

Q48. (ARITHMETIC - MEN-DAYS WORK EQUIVALENCE) If 12 men or 18 women can finish a piece of work in 14 days, in how many days can 8 men and 16 women do it?

- A) 9 days
- B) 10 days
- C) 8 days
- D) 12 days

✓ **Correct Answer: Option A**
Work rate equivalence: 12 Men = 18 Women $\Rightarrow$ 1 Man = 1.5 Women.
8 Men + 16 Women = $(8 \times 1.5) + 16 = 12 + 16 = 28$ Women.
Using $M_1D_1 = M_2D_2$: $18 \times 14 = 28 \times D \Rightarrow D = (18 \times 14) / 28 = 9$ days.

DATA SET (Q49 - Q52): PRODUCT SALES PERFORMANCE
The table below shows annual sales revenue (₹ in Lakhs) of 4 product lines (P, Q, R, S) across 3 years:

Product Line	2023	2024	2025
Product P	150	180	225
Product Q	200	220	242
Product R	100	130	169
Product S	250	275	300

Q49. (DATA INTERPRETATION - PERCENTAGE GROWTH) Which product line recorded the highest percentage growth in sales revenue between 2023 and 2025?

- A) Product P
- B) Product Q
- C) Product R
- D) Product S

✓ **Correct Answer: Option C**
Percentage growth from 2023 to 2025:
• Product P: $(225 - 150)/150 = 75/150 = 50.0\%$
• Product Q: $(242 - 200)/200 = 42/200 = 21.0\%$
• Product R: $(169 - 100)/100 = 69/100 = 69.0\%$
• Product S: $(300 - 250)/250 = 50/250 = 20.0\%$
Product R recorded the highest percentage growth at 69.0%.

Product Line	2023	2024	2025
Product P	150	180	225
Product Q	200	220	242
Product R	100	130	169
Product S	250	275	300

Q50. (DATA INTERPRETATION - SUMMATION & TABLE READING) What is the total sales revenue (₹ in Lakhs) generated across all 4 product lines in 2024?

A) 805

B) 800

C) 815

D) 790

✓ **Correct Answer: Option A**

Total 2024 Revenue = $180 + 220 + 130 + 275 = ₹ 805$ Lakhs.

Product Line	2023	2024	2025
Product P	150	180	225
Product Q	200	220	242
Product R	100	130	169
Product S	250	275	300

Q51. (DATA INTERPRETATION - RATIOS) What is the ratio of Product P's sales in 2023 to Product S's sales in 2025?

A) 1 : 2

B) 3 : 4

C) 2 : 3

D) 1 : 1

✓ **Correct Answer: Option A**

Ratio = Sales of P (2023) : Sales of S (2025) = $150 : 300 = 1 : 2$.

Product Line	2023	2024	2025
Product P	150	180	225
Product Q	200	220	242
Product R	100	130	169
Product S	250	275	300

Q52. (DATA INTERPRETATION - PERCENTAGE SHARE) Product R's sales in 2025 constitute approximately what percentage of total company sales in 2025?

A) 18.0%

B) 18.1%

C) 20.0%

D) 16.5%

✓ **Correct Answer: Option B**

Total sales in 2025 = 225 + 242 + 169 + 300 = 936 Lakhs.

Percentage for Product R = $(169 / 936) \times 100 = 18.055\% \approx 18.1\%$.

Q53. (DATA SUFFICIENCY - NUMBER PROPERTIES) Data Sufficiency: Is positive integer x even? Statement I: $x + 3$ is an odd integer. Statement II: $3x$ is an even integer.

A) Statement I alone is sufficient.

B) Statement II alone is sufficient.

C) BOTH statements TOGETHER are necessary.

D) EITHER statement ALONE is sufficient.

✓ **Correct Answer: Option D**

• Statement I: $x + 3 = \text{odd} \Rightarrow x = \text{odd} - 3 = \text{even}$. (Sufficient alone)

• Statement II: $3x = \text{even} \Rightarrow$ since 3 is odd, x must be even. (Sufficient alone)

Hence, EITHER statement ALONE is sufficient.

Q54. (DATA SUFFICIENCY - SPEED & DISTANCE) Data Sufficiency: What is the value of speed of a train? Statement I: Train crosses a telegraph post in 10 seconds. Statement II: Length of the train is 150 meters.

A) Statement I alone is sufficient.

B) Statement II alone is sufficient.

C) BOTH statements TOGETHER are necessary.

D) Statements I and II together are NOT sufficient.

✓ **Correct Answer: Option C**

Speed = Length of Train / Time to cross stationary point.

Statement I gives Time = 10s (insufficient alone).

Statement II gives Length = 150m (insufficient alone).

Combining both gives Speed = $150 / 10 = 15$ m/s. BOTH statements TOGETHER are necessary.

Q55. (DATA SUFFICIENCY - GEOMETRY & MENSURATION) Data Sufficiency: What is the area of rectangle ABCD?

Statement I: Diagonal of rectangle is 10 cm. Statement II: Perimeter of rectangle is 28 cm.

A) Statement I alone is sufficient.

B) Statement II alone is sufficient.

C) BOTH statements TOGETHER are necessary.

D) EITHER statement ALONE is sufficient.

✓ **Correct Answer: Option C**

• From Statement I: $l^2 + w^2 = 10^2 = 100$.

• From Statement II: $2(l + w) = 28 \Rightarrow l + w = 14$.

Using identity $(l + w)^2 = l^2 + w^2 + 2lw \Rightarrow 14^2 = 100 + 2lw \Rightarrow 196 - 100 = 2lw \Rightarrow 2lw = 96 \Rightarrow lw = 48$ cm².

BOTH statements TOGETHER are necessary.

Q56. (DATA SUFFICIENCY - INEQUALITIES) Data Sufficiency: Is $a > b$? Statement I: $a - b > 0$. Statement II: $a^2 > b^2$.

A) Statement I alone is sufficient.

B) Statement II alone is sufficient.

C) BOTH statements TOGETHER are necessary.

D) Statements I and II together are NOT sufficient.

✓ **Correct Answer: Option A**

• Statement I: $a - b > 0 \Rightarrow a > b$. This directly answers the question affirmatively. (Sufficient alone)

• Statement II: $a^2 > b^2$ could mean $a = -5$, $b = 2$ where $a^2 > b^2$ but $a < b$. (Insufficient alone)

Therefore, Statement I alone is sufficient.

Q57. (ALGEBRA - EXPONENTS & INDICES) Solve for x : $2^{x+2} + 2^x = 40$.

A) 3

B) 4

C) 2

D) 5

✓ **Correct Answer: Option A**

$2^x \times 2^2 + 2^x = 40 \Rightarrow 2^x (4 + 1) = 40 \Rightarrow 5 \times 2^x = 40 \Rightarrow 2^x = 8 \Rightarrow x = 3$.

Q58. (ALGEBRA - IDENTITIES) If $a + b = 8$ and $ab = 15$, find $a^2 + b^2$.

A) 34

B) 49

C) 30

D) 28

✓ **Correct Answer: Option A**

$a^2 + b^2 = (a + b)^2 - 2ab = 8^2 - 2(15) = 64 - 30 = 34$.

Q59. (ALGEBRA - QUADRATIC EQUATIONS) Find roots of the quadratic equation: $x^2 - 7x + 12 = 0$.

A) 3, 4

B) -3, -4

C) 2, 5

D) 1, 6

✓ **Correct Answer: Option A**

$$x^2 - 3x - 4x + 12 = 0 \Rightarrow (x - 3)(x - 4) = 0 \Rightarrow x = 3, 4.$$

Q60. (ALGEBRA - LOGARITHMS) Value of $\log_3(81) + \log_2(32) = ?$

A) 9

B) 8

C) 7

D) 10

✓ **Correct Answer: Option A**

$$\log_3(81) = \log_3(3^4) = 4.$$

$$\log_2(32) = \log_2(2^5) = 5.$$

$$\text{Sum} = 4 + 5 = 9.$$

Q61. (ALGEBRA - ARITHMETIC PROGRESSIONS) Sum of first 20 terms of AP: 5, 9, 13, 17 ... is:

A) 860

B) 800

C) 900

D) 830

✓ **Correct Answer: Option A**

First term $a = 5$, common difference $d = 4$, $n = 20$.

$$S_{20} = (n/2)[2a + (n - 1)d] = (20/2)[2(5) + 19(4)] = 10[10 + 76] = 10(86) = 860.$$

Q62. (GEOMETRY - PYTHAGORAS THEOREM) In a right triangle, base = 6 cm, height = 8 cm. Find hypotenuse length.

A) 10 cm

B) 12 cm

C) 14 cm

D) 9 cm

✓ **Correct Answer: Option A**

$$\text{Hypotenuse } h = \sqrt{(\text{base}^2 + \text{height}^2)} = \sqrt{(6^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10 \text{ cm}.$$

Q63. (GEOMETRY - CIRCLES & MENSURATION) Area of a circle is 154 cm^2 . Find its circumference (Use $\pi = 22/7$).

A) 44 cm

B) 22 cm

C) 88 cm

D) 66 cm

✓ **Correct Answer: Option A**

$$\text{Area} = \pi r^2 = 154 \Rightarrow (22/7) r^2 = 154 \Rightarrow r^2 = 49 \Rightarrow r = 7 \text{ cm}.$$

$$\text{Circumference} = 2\pi r = 2 \times (22/7) \times 7 = 44 \text{ cm}.$$

Q64. (MENSURATION - 3D SOLIDS) Find volume of a cylinder with radius 7 cm and height 10 cm.

A) 1,540 cm³

B) 1,450 cm³

C) 1,200 cm³

D) 1,600 cm³

✓ **Correct Answer: Option A**

Volume = $\pi r^2 h = (22/7) \times 7^2 \times 10 = (22/7) \times 49 \times 10 = 22 \times 7 \times 10 = 1,540 \text{ cm}^3$.

Q65. (MODERN MATH - PERMUTATIONS) In how many ways can 5 distinct books be arranged on a shelf?

A) 120

B) 60

C) 24

D) 720

✓ **Correct Answer: Option A**

Number of arrangements = $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ ways.

Q66. (MODERN MATH - COMBINATIONS) Value of 7C_3 is:

A) 35

B) 21

C) 42

D) 30

✓ **Correct Answer: Option A**

${}^7C_3 = (7 \times 6 \times 5) / (3 \times 2 \times 1) = 210 / 6 = 35$.

Q67. (MODERN MATH - PROBABILITY) Probability of drawing an Ace from a well-shuffled standard deck of 52 cards:

A) 1/13

B) 1/52

C) 1/4

D) 4/13

✓ **Correct Answer: Option A**

Total cards = 52. Number of Aces = 4.

Probability = $4 / 52 = 1 / 13$.

Q68. (MODERN MATH - PROBABILITY) Two dice are rolled. Probability that the sum of face numbers is 7:

A) 1/6

B) 1/12

C) 5/36

D) 1/4

✓ **Correct Answer: Option A**

Total outcomes = $6 \times 6 = 36$.

Favorable pairs with sum 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) = 6 pairs.

Probability = $6 / 36 = 1/6$.

Q69. (ALGEBRA - GEOMETRIC PROGRESSIONS) Find common ratio of GP: 3, 6, 12, 24 ...

A) 2

B) 3

C) 0.5

D) 4

✓ **Correct Answer: Option A**

Common ratio $r = 6 / 3 = 12 / 6 = 2$.

Q70. (GEOMETRY - QUADRILATERALS) Perimeter of a square is 48 cm. Find its area.

A) 144 cm²

B) 121 cm²

C) 169 cm²

D) 100 cm²

✓ **Correct Answer: Option A**

Side of square $s = 48 / 4 = 12$ cm.

Area = $s^2 = 12^2 = 144$ cm².

Q71. (ALGEBRA - INEQUALITIES) Solve inequality: $3x - 5 < 10$.

A) $x < 5$

B) $x > 5$

C) $x < 3$

D) $x < 15$

✓ **Correct Answer: Option A**

$3x - 5 < 10 \Rightarrow 3x < 15 \Rightarrow x < 5$.

Q72. (GEOMETRY - COORDINATE GEOMETRY) Distance between points (2, 3) and (5, 7) in Cartesian plane:

A) 5

B) 6

C) 7

D) 4

✓ **Correct Answer: Option A**

Distance $d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]} = \sqrt{[(5 - 2)^2 + (7 - 3)^2]} = \sqrt{[3^2 + 4^2]} = \sqrt{[9 + 16]} = \sqrt{25} = 5$.